

Survivorship Validation

Testing The Edge against real point-in-time Russell 1000 data

Prepared 22 July 2026 • Norgate Data trial (2-year window) • Internal research note

Bottom line. We built the tooling to test whether The Edge's track record depends on survivorship bias, and ran it. **The test came back inconclusive.** The two-year data trial is too short to answer the question: the result swings between -2% and $+55\%$ a year depending purely on which day of the month you start, which is far larger than the effect we were trying to measure. An earlier version of this analysis reported a clean pass ($+28.5\%/yr$). **That result was wrong and has been retracted** — a code review found a look-ahead error and a lucky start date. This report documents both the finding and the error, because the error is the more useful lesson.

1. The question

Survivorship bias is the most common way a backtest lies to you. Our stock universe is built from companies that exist *today*. That quietly excludes every company that went bankrupt, got acquired, or was thrown out of the index along the way. If your strategy would have bought some of those losers, a backtest built only on today's survivors never shows the damage — so it looks better than reality.

Our universe is missing roughly 89% of delisted names. So the question was simple and important: **is The Edge's performance real, or is it an artifact of only ever seeing the winners?**

2. What we built

Norgate Data sells point-in-time index data: it knows exactly who was in the Russell 1000 on any given date, and it keeps pricing dead companies right through their delisting. We built two pieces of tooling against it:

- **norgate_ingest.py** — pulls every company that was in the Russell 1000 at any point in the window, along with its total-return prices and its day-by-day index membership flag.
- **norgate_devalidate.py** — re-runs The Edge on that universe. At each rebalance it can only pick from companies that were *actually in the index that day*, and if a holding delists mid-period it is priced through the failure rather than quietly dropped.

What the data gave us:

Measure	Value	Note
Companies pulled	1,135	Russell 1000 current <i>and</i> past
Of which delisted	50	the names a biased universe hides
Window	17 Jul 2024 – 16 Jul 2026	hard cap of the free trial
Trading days	501	~2 years

Rebalances per sleeve	9	2-month holding clock
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That last row is the whole problem, and it is worth pausing on. Nine decisions is not a sample you can conclude anything from. We did not know how badly that would bite until we measured it.

3. The first answer — and why it was wrong

The first run looked like unambiguous good news. On the real, de-biased universe The Edge returned **28.5%/yr against the S&P's 16.3%**, with a shallower drawdown than our normal backtest. Read at face value: removing the bias made the model look *better*, so survivorship was never flattering us.

A code review of that script found three separate errors, which compounded in the same flattering direction:

Error	What it did	Effect
Look-ahead	The model read a day's closing price to pick stocks, then bought at that <i>same</i> closing price. In reality you see the close and can only trade the next day.	Free profit on every trade
Lucky start date	The test started on one arbitrary day. We never checked whether a different start date gave a different answer.	Cherry-picked a good draw
Drawdown mismeasured	Losses were sampled once every 42 days, then compared against our normal backtest's daily measurement. Crashes that happened and recovered between samples were invisible.	Hid the real risk

Why this matters more than the result. None of these were exotic. Each one pushed the answer in the direction we were hoping for, which is exactly why none of them looked suspicious at the time. The number was only caught because the code was reviewed *after* it produced a pleasing answer — not before.

4. The corrected result

We rebuilt the test to match how the product actually trades: buy the day *after* the signal, run two staggered sleeves, check the 200-day market-regime rule every day, and measure losses daily.

	Return/yr	Sharpe	Worst drop
The Edge (de-biased, corrected)	3.3%	0.14	−36.2%
S&P 500 (total return)	18.9%	0.86	−18.7%
<i>Original (retracted)</i>	28.5%	0.95	−11%

Taken alone, that top row reads like a disaster. It isn't — and this is the genuinely important part of the report. That 3.3% is **also** just one draw. When we re-ran the same test starting on 14 different days of the month, we got this:

Start offset (days)	0	3	6	9	12	15	18
Return/yr	7.6%	33.1%	6.7%	6.0%	5.3%	24.6%	−2.4%
Start offset (days)	21	24	27	30	33	36	39
Return/yr	1.8%	13.6%	44.0%	51.6%	54.8%	21.0%	38.0%

Same strategy. Same data. Same two years. Returns from –2.4% to +54.8%, purely from which day you happen to start. The median is 17.3%, sitting right on top of the S&P's 18.9%; exactly half the start dates (7 of 14) beat the market.

The honest conclusion. Over a two-year window with nine decisions, this test is a coin flip. The noise is many times larger than the survivorship effect we were trying to detect. It cannot confirm the bias is harmless, and it cannot prove it is fatal. **Survivorship stays an open question.** Anyone quoting either the 28.5% or the 3.3% as a finding is quoting noise.

5. A second correction: the rigor claims on the site

While validating the above, we re-ran the overfitting statistics published on the */model* page and found them stale and flattering. They have been corrected.

Statistic	Page claimed	Actual	Status
Configurations screened	~150	120	corrected
Significance (Newey-West t)	~2.8	+2.17	still significant
Deflated Sharpe	>0.95	~1.00	was correct
Overfitting probability (PBO)	~0.27	~0.5, unstable	materially wrong

What PBO is, in plain language. It answers: *when I picked my best settings, was I finding real signal or just picking noise?* The test splits the history in half every possible way, finds the settings that looked best on one half, then checks how those same settings actually ranked on the other half. If the winner routinely turns out mediocre later, you were fitting noise.

- **Near 0** — the winner keeps winning on fresh data. Healthy.
- **Near 0.5** — picking the winner is a coin flip. You are selecting noise.
- **Above 0.5** — the winner is reliably *worse* later. Actively misleading.

The page claimed 0.27 (comfortably healthy). The real figure is about 0.5 — and it is **not stable**: the identical test measured 0.67 on one data vintage and 0.49 a week later. Publishing the 0.49 alone would have been picking the flattering sample, which is precisely the bias PBO exists to detect. The page now reports it as inconclusive and shows both figures.

Important nuance. A PBO near 0.5 does *not* mean the acceleration signal is fake. It means the *dials* — basket size and growth mix — cannot be reliably optimised, because thirty near-identical settings all perform about the same, so there is no persistent winner to find. The real defence against dial-mining is that we never picked the winner: the shipped configuration ranks **59th of 120**, chosen on economic reasoning, not on backtest rank.

6. Where this leaves the product

- **Nothing about the live model changed.** No signal, weight, or rule was altered by this work. What changed is what we are entitled to claim about it.

- **Survivorship is unresolved**, not cleared. The /model page has been reverted to say so, and the earlier “the edge survives” language has been removed.
- **The tooling is built and works**. Point it at a full-history dataset and it answers the question properly — the blocker is data, not code.
- **Two published statistics were corrected** in the direction of being less flattering.
- **The strongest honest claim remains unchanged** from earlier work: sell the risk-adjusted shape and drawdown control, not secret alpha. Roughly 70% of the return is buyable factor premium.

7. What it would take to actually settle it

The trial caps history at two years, which is the single binding constraint. A paid full-history point-in-time licence (Norgate Platinum) would give roughly twenty years — on the order of 120 rebalances per sleeve instead of 9. At that length the start-date noise averages out and the survivorship effect becomes measurable. The same two scripts run unchanged against it.

One further limitation to carry forward: the Norgate stock package contains no fundamentals, so this test necessarily ran the model in pure-momentum mode with the revenue-growth blend switched off. Even with full history, validating the shipped 75% growth mix needs a fundamentals source joined to the same point-in-time universe.

8. Reproducibility

Everything in this report regenerates from the repository:

File	Purpose
norgate_ingest.py	Pulls the point-in-time universe; caches to data/cache/
norgate_devalidate.py	Runs the de-biased test + the start-date sweep
edge_rigor_honest.py	Regenerates Newey-West t, Deflated Sharpe, PBO
commit 19eb37c	The retraction and corrected basis

Figures quoted: de-biased book 3.3%/yr, Sharpe 0.14, max drawdown –36.2% over 374 trading days (9 rebalances per sleeve, two sleeves); S&P 500 total return 18.9%, Sharpe 0.86, –18.7% over the same span; start-offset sweep across 14 phases, median 17.3%, range –2.4% to +54.8%. Universe 1,135 names including 50 delisted, 17 Jul 2024 to 16 Jul 2026.